

AI Developer Candidate Assignment

Travel Reimbursement Approval Agent

Purpose: Assess hands-on ability to build a practical GenAI and Agentic AI solution, not just describe one.

Business scenario	Review employee travel reimbursement claims against policy, receipts, limits, and approval rules.
Build objective	Create a working AI-assisted agent that evaluates a claim and returns a structured recommendation.
Expected effort	Designed to be completed in 2-3 days using free-tier, open-source, or local tools.
Expected outcome	Runnable prototype, sample inputs/outputs, and a short explanation of design choices and trade-offs.

1. Assignment Overview

Build a working prototype of a Travel Reimbursement Approval Agent. The agent should accept a claim, use policy/rule context and tools, and produce a clear decision such as Approve, Partially Approve, Reject, or Manual Review. The emphasis is on clean implementation, GenAI reasoning, tool usage, and reliability in ambiguous cases.

2. Scope and Constraints

- Keep the solution practical and lightweight; do not build a production-grade enterprise system.
- Use Python, JavaScript/TypeScript, or another language you are comfortable with.
- You may use LangChain, LlamaIndex, Semantic Kernel, CrewAI, AutoGen, direct LLM APIs, or an equivalent lightweight approach.
- Use the policy in Appendix A and the sample claims in Appendix B (below); do not invent your own claim set. No real employee or company data should be used.
- Deliver your solution as a single Jupyter notebook so the demo is easy to run and review end-to-end.

3. Minimum Functional Expectations

- **Claim intake:** Accept a reimbursement claim from JSON, CSV, form input, or API request.
- **Context grounding:** Retrieve or reference relevant policy/rule context before making a decision.
- **Tool/function usage:** Use at least two meaningful tools or functions, such as policy lookup, receipt completeness check, per-diem/limit checker, duplicate detector, approval threshold check, or output validator.
- **GenAI/Agentic workflow:** Show how the LLM decides when to use tools, combines results, and handles missing or conflicting information.
- **Structured output:** In the final code cell, output a JSON array with one object per provided claim, each containing exactly: claim_id, decision (one of APPROVE, PARTIAL_APPROVE, REJECT, MANUAL_REVIEW), approved_amount, deducted_amount, missing_docs, policy_refs, confidence, explanation, and tools_used.
- **Manual review handling:** Route uncertain, incomplete, or policy-exception cases to Manual Review rather than forcing a decision.
- **Results dashboard:** Include a minimal, data-driven dashboard summarizing your results (for example, a decision breakdown or approved vs deducted amounts), derived from the actual claim outcomes.

4. Provided Inputs

- The travel reimbursement policy — see Appendix A. Every rule has a stable id (POL-*) you can cite in your output.
- A small limit table, approval matrix, or category eligibility file.
- The five sample claims to evaluate — see Appendix B. Return one result object for each of the five claims.
- Optional mock receipts or receipt metadata, such as date, amount, category, vendor, and attachment status.

5. Deliverables

- **Runnable notebook:** A single Jupyter notebook named yourname.ipynb that runs top-to-bottom without manual steps.
- **Structured results cell:** The final code cell outputs the JSON results for all provided claims (fields as listed in section 3).
- **Design notes & reasoning:** A markdown section titled “Design Notes & Reasoning” covering your assumptions, trade-offs, why you route certain cases to Manual Review, and what you would improve next.

- **User Interface** : A minimal results interface, saved as UI SS_1.png (optionally UI SS_2.png) next to the notebook, or rendered inline under a “## Dashboard” heading. This UI will be the means by which the user interacts with notebook code.
 - **README**: Setup steps, required environment variables, how to run the demo, and key design choices. (should be part of notebook only)
 - **Sample outputs**: At least three example claims with generated decisions and explanations as part of the notebook
 - **Demo evidence**: Screenshots, API responses, notebook output, or a short walkthrough. **Be prepared to demonstrate the prototype during the interview.**
- Assumptions and limitations**: Briefly list assumptions, simplifications, known gaps, and what you would improve next.

NOTE : Single Notebook uploaded on github with working repo (Shareable link) is the only deliverable (namesurname.ipynb) we will assess

6. Evaluation Criteria

- Hands-on implementation quality: runnable, understandable, and reasonably organized code.
- Practical use of GenAI and Agentic AI: tool calling, workflow control, prompting, and context handling.
- Business correctness: sensible reimbursement decisions grounded in policy/rule context.
- Reliability: structured outputs, validation, fallbacks, and manual review for uncertain cases.
- Developer judgement: clear trade-offs, simple design, and avoidable over-engineering removed.

7. Optional Enhancements

- Simple UI or chatbot interface.
- Audit trail showing retrieved context, tools called, and intermediate checks.
- Basic test cases or evaluation script for sample claims.
- Confidence score or reason codes for manual review.
- MCP-based tool integration, if you are comfortable implementing it within the timebox.

Submission guidance: This assignment is intentionally timeboxed to 2-3 days. A small, working, well-explained prototype is preferred over a broad but fragile implementation.

Appendix A — Travel Reimbursement Policy

Mock policy for this assignment. No real company or employee data. Every rule has a stable id (POL-) you should cite in your output where relevant.*

1. Eligible & Ineligible Categories

POL-CAT-01 — Eligible categories — Reimbursable when incurred for a documented business purpose:

- Airfare (economy class only — see POL-AIR-01)
- Lodging (hotel room charges)
- Meals (subject to per-diem limits — see POL-PD-01)
- Ground transport (taxi, rideshare, train, rental car, parking)
- Conference / registration fees

POL-CAT-02 — Ineligible items — Never reimbursable; rejected (deducted in full):

- Alcohol and minibar charges
- Spa, gym, and personal entertainment
- In-room movies, personal shopping, gifts
- Traffic fines, penalties, and late fees
- Any personal (non-business) expense

2. Per-Diem & Category Limits

POL-PD-01 — Meals — Maximum \$75 per day. Amounts above the daily cap are deducted; the rest is reimbursed.

POL-PD-02 — Lodging — Maximum \$200 per night. Amounts above the nightly cap are deducted; the rest is reimbursed.

POL-PD-03 — Ground transport — Maximum \$50 per day. Amounts above the cap are deducted.

POL-AIR-01 — Airfare class — Only economy class airfare is reimbursable. Business/first-class fares are a policy exception and must be routed to Manual Review (not auto-deducted, because a pre-approval may exist).

3. Receipt Rules

POL-RCT-01 — Receipt required above \$25 — Any single line item greater than \$25 requires an attached, itemized receipt. Airfare and lodging always require a receipt regardless of amount.

POL-RCT-02 — Missing receipt handling — If a receipt is missing for an item that requires one, the item is not silently rejected — the claim is routed to Manual Review so the reviewer can request the receipt.

4. Approval Thresholds

Thresholds are evaluated on the total reimbursable amount (after per-diem deductions but before final decision).

POL-APR-01 — Auto-approve tier — Total ≤ \$500: may be auto-approved by the agent if fully compliant.

POL-APR-02 — Manager tier — Total > \$500 and ≤ \$2,000: eligible for approval, treated as approvable when fully compliant.

POL-APR-03 — Director / Manual-Review tier — Total > \$2,000: exceeds the agent's auto-approval authority and must be routed to Manual Review (director approval required), even if otherwise compliant.

5. Timeliness

POL-TIME-01 — Submission window — Claims must be submitted within 30 days of the expense date. Late claims are routed to Manual Review.

Decision Guidance (how the rules combine)

Approve — every item eligible, all receipts present, all within per-diem, total within an approvable tier (POL-APR-01 / POL-APR-02).

Partially Approve — the claim is valid but some amounts exceed per-diem caps; reimburse up to the cap and deduct the excess.

Reject — the claimed items are ineligible (POL-CAT-02) with nothing reimbursable.

Manual Review — any ambiguity, policy exception, high value (POL-APR-03), missing required receipt (POL-RCT-02), or conflicting information. Prefer Manual Review over forcing a decision.

Appendix B — Sample Claims to Evaluate

Evaluate all 5 claims below against the policy in Appendix A. Return one result object per claim using the fields defined in Section 3. Currency: USD.

CLM-001 — Attend 2-day industry conference (business)

Employee: A. Rivera · Trip: 2026-06-10 to 2026-06-12 · Submitted: 2026-06-20

Category	Description	Amount (USD)	Receipt attached
airfare	Round-trip economy airfare	420.00	Yes
lodging	Hotel, 2 nights @ \$180	360.00	Yes
meals	Meals, 3 days @ ~\$60/day	180.00	Yes
conference_fees	Conference registration	150.00	Yes

Total claimed: \$1110.00

CLM-002 — Weekend hotel stay

Employee: B. Osei · Trip: 2026-06-14 to 2026-06-15 · Submitted: 2026-06-25

Category	Description	Amount (USD)	Receipt attached
spa	Hotel spa package	300.00	Yes
minibar	In-room minibar	80.00	Yes

Total claimed: \$380.00

CLM-003 — Client site visit (business)

Employee: C. Nakamura · Trip: 2026-06-08 to 2026-06-10 · Submitted: 2026-06-22

Category	Description	Amount (USD)	Receipt attached
airfare	Round-trip economy airfare	300.00	Yes
lodging	Hotel, 2 nights @ \$250	500.00	Yes
meals	Meals, 2 days @ \$70/day	140.00	Yes

Total claimed: \$940.00

CLM-004 — International vendor negotiation (business)

Employee: D. Fischer · Trip: 2026-06-16 to 2026-06-18 · Submitted: 2026-06-28

Category	Description	Amount (USD)	Receipt attached
airfare	Business-class international airfare	2400.00	Yes
lodging	Hotel, 3 nights	600.00	No

Total claimed: \$3000.00

CLM-005 — Client dinner / business development

Employee: E. Haddad · Trip: 2026-06-11 to 2026-06-11 · Submitted: 2026-06-24

Category	Description	Amount (USD)	Receipt attached
meals	Client dinner for 4 (business development)	220.00	No

Total claimed: \$220.00